

pre-treatment violation of parallel trends. As a sensitivity analysis, the researcher might also report confidence sets that allow the maximum post-treatment violation of parallel trends to be up to $\bar{M}$ times larger than the maximum pre-treatment violation for different values of $\bar{M}$. The proposed sensitivity analysis thus follows the “top-down” approach described by Tamer (2010), in which the researcher reports what can be learned under a sequence of progressively weaker assumptions. Performing such sensitivity analyses makes clear what must be assumed about the possible differences in trends in order to draw specific causal conclusions, in line with the partial identification paradigm initiated by Charles Manski and co-authors in a variety of contexts (Manski, 1989, 1990, 2003, 2008). We provide the HonestDiD R and Stata packages to implement our methods.¹ We illustrate our recommended approach with applications to two recently published papers, in which we show how the choice of the restrictions Δ can be tailored to the economic context.

1.1. Related literature

The approach in this paper builds on the foundational partial identification analysis for DiD in Manski and Pepper (2018). Manski and Pepper consider identification under researcher-specified bounds on the magnitude of δ_{post} (what they call “bounded DiD variation”), and calibrate these bounds using the maximal pre-treatment violation of parallel trends in their empirical application on the effects of right-to-carry gun laws.² One of our leading classes of restrictions, Δ^{RM} , formalizes this calibration approach by bounding the magnitude of post-treatment violations of parallel trends by $\bar{M}$ times the maximal pre-treatment violation. Our framework also allows for many other intuitive restrictions—such as bounds on how far δ can deviate from linearity—and it can be applied to a variety of DiD estimators, including recent proposals for settings with staggered treatment timing. Most importantly, while Manski and Pepper (2018) provide a framework for identification, we provide inference methods to construct uniformly valid confidence sets for the treatment effect of interest. This allows applied researchers to account for statistical uncertainty in their analyses, which can be important since event-study coefficients are often imprecisely estimated in practice.

Several other recent papers consider various relaxations of the parallel trends assumption. Keele *et al.* (2019) develop techniques for testing the sensitivity of DiD designs to violations of the parallel trends assumption, but they do not incorporate information from the observed pre-trends in their sensitivity analysis. Empirical researchers commonly adjust for the extrapolation of a linear trend from the pre-treatment periods when there are concerns about violations of the parallel trends assumption, which is valid if the difference in trends is exactly linear (*e.g.* Bhuller *et al.*, 2013; Dobkin *et al.*, 2018; Goodman-Bacon, 2018, 2021). Our methods nest this approach as a special case, but allow for valid inference under less restrictive assumptions about the class of possible differences in trends (such as when δ is only approximately linear). Freyaldenhoven *et al.* (2019) propose a method that allows for violations of the parallel trends assumption but requires an additional covariate that is affected by the same confounding factors as the outcome but not by the treatment of interest. Ye *et al.* (2020) consider partial identification of treatment effects when there exist two control groups whose outcomes have a bracketing relationship with

1. The latest version of the R and Stata packages are respectively available at <https://github.com/asheshrambachan/HonestDiD> and <https://github.com/mcaceresb/stata-honestdid/>.

2. Manski and Pepper (2018) also consider “bounded time” and “bounded state” restrictions that bound how much the mean of $Y(0)$ can differ either across treatment groups or within-groups over time. Such restrictions could also be incorporated into our framework by augmenting the vector $\hat{\beta}$ to include group-specific sample averages.

the outcome of the treated group. [Leavitt \(2020\)](#) proposes an empirical Bayes approach calibrated to pre-treatment differences in trends, and [Bilinski and Hatfield \(2020\)](#) and [Dette and Schumann \(2020\)](#) propose non-inferiority approaches based on pre-tests for the magnitude of the pre-treatment violations of parallel trends.

Our methods address several concerns related to current empirical practice in DiD and event-study designs. First, pre-trends tests may be underpowered against meaningful violations of parallel trends, potentially leading to severe undercoverage of conventional confidence intervals ([Freyaldenhoven et al., 2019](#); [Bilinski and Hatfield, 2020](#); [Kahn-Lang and Lang, 2020](#); [Roth, 2022](#)). Second, statistical distortions from pre-trends tests may further undermine the performance of conventional inference procedures ([Roth, 2022](#)). Third, parametric approaches to controlling for pre-existing trends may be sensitive to functional form assumptions ([Wolfers, 2006](#); [Lee and Solon, 2011](#)). We address these issues by providing tools for inference that do not rely on an exact parallel trends assumption, incorporate statistical uncertainty about the estimated event-study coefficients, and make clear the mapping between the researcher's assumptions about the potential differences in trends and the strength of their causal conclusions.

Our work complements a growing literature on the causal interpretation of event-study coefficients in two-way fixed effects models in the presence of staggered treatment timing and heterogeneous treatment effects ([Borusyak and Jaravel, 2016](#); [Callaway and Sant'Anna, 2020](#); [de Chaisemartin and D'Haultfœuille, 2020](#); [Sun and Abraham, 2020](#); [Athey and Imbens, 2021](#); [Goodman-Bacon, 2021](#)). A key finding is that regression coefficients from conventional approaches may not produce convex weighted averages of treatment effects even if parallel trends holds. Several alternative estimators have been proposed that consistently estimate interpretable causal estimands under a suitable parallel trends assumption. Our methodology can be used in conjunction with these alternative estimators to assess their sensitivity to violations of the corresponding parallel trends assumption; see [Section 2.1](#) for additional details.

More broadly, our work contributes to a larger econometric literature that uses partial identification to provide empirical researchers with tractable tools to conduct inference under assumptions that may be more credible in empirical practice; see, for example, [Manski \(2003, 2007, 2013\)](#), [Tamer \(2010\)](#), [Ho and Rosen \(2017\)](#), and [Molinari \(2020\)](#) for reviews.

2. MODEL SET-UP

2.1. Event-study coefficients

We suppose that the researcher has estimated a vector of “event-study coefficients” $\hat{\beta}_n \in \mathbb{R}^{T+\bar{T}}$, which can be partitioned into vectors of coefficients corresponding with the pre-treatment and post-treatment periods, $\hat{\beta}_n = (\hat{\beta}'_{n,pre}, \hat{\beta}'_{n,post})$, where $\hat{\beta}_{n,pre} \in \mathbb{R}^T$ and $\hat{\beta}_{n,post} \in \mathbb{R}^{\bar{T}}$. Event-study estimates of this form arise from non-staggered DiD as well as a variety of related estimators, as we illustrate with several examples

Example 1 (Non-staggered DiD). Consider the canonical DiD setting in which we have a balanced panel of units from period $t = -\bar{T}, \dots, \bar{T}$, and units with $D_i = 1$ receive a treatment beginning in period $t = 1$, while units with $D_i = 0$ never receive the treatment. It is common to report DiD estimates of the form

$$\hat{\beta}_s = (\bar{Y}_{s1} - \bar{Y}_{s0}) - (\bar{Y}_{01} - \bar{Y}_{00}),$$

where $\bar{Y}_{sd}$ is the sample mean of the outcome for units with $D_i = d$ in period $t = s$. Intuitively, $\hat{\beta}_s$ compares the change in the mean outcome between period 0 and period s for the treated and comparison units. In this setting, the estimates $\hat{\beta}_s$ are numerically equivalent to the OLS